King Abdulaziz University



URP 233

Urban Regeneration

Supervisor : Prof .D. Mohamed Maatouk Prof . D. Allock Tawari Ram Tawari Dr abdullah mahboob

Basem Al-Hussain Haij
Firas Hazem Alharbi
Faisal Nasser Alamri
Abdullah Khaled Samman
Abdullah Hussein Radwan
Aseel Fadil Kuna
Meshari Abdulrhman Al-shehri
Mohammed Abdulmohsin Alshammari
Sameer Saeed Bahamdon

Table of Contents

Urban Regeneration

01 **Theoretical Background**

-Introduction to Al-Ruwais District

-Theoretical Background of Urban Regeneration

Page 3 - 4

02 **Best Practice**

-NewYork City (Lower Manhattan)
-London
-Melbourne

Page 5 - 8

03 **Study of the Current Situation of Al-Ruwais Districts**

-Introduction
-Current Situation Analysis of Al-Ruwais District
-Current Transportation Situation Analysis in Al-Ruwais District

Page 8 - 13

04 **General Site Analysis**

-Urban Analysis
-Satisfaction Rates of Services
-Transportation Analysis

Page 14 - 17

URBAN REGENERATION

Origin of the Idea

Urban renewal is a planned process aimed at rehabilitating and developing deteriorated urban fabrics by improving infrastructure, buildings, and public services, as well as revitalizing economic and social functions. This process seeks to achieve urban sustainability and enhance the quality of life in cities, while taking into account the cultural and environmental dimensions of the place.

Key Characteristics of Urban Renewal

- **Comprehensive scope**
- **Variety of tools**
- **Long-term planning**
- **Participatory approach**
- **Focus on identity**

Historical Stages of Urban Renewal

A) Reconstruction Phase (1945–1960)

- **Focus: Rebuilding war-damaged cities.**
- **Features: Large-scale housing projects, infrastructure modernization, construction of highways.**
- **Advantage: Rapid revival of cities.**
- **Disadvantage: Emphasis on physical rebuilding with little attention to social aspects.**

B) Urban Renewal Phase (1960–1970)

- **Focus: Demolition of informal settlements and construction of modern housing complexes.**
- **Features: Emergence of large-scale housing projects (high-rise housing).**
- **Advantage: Improvement of the residential environment and provision of new public facilities.**
- **Disadvantage: Led to social isolation and decline in quality of life due to weakened community ties.**

c) Economic-Social Renewal Phase (1970–1990)

- **Focus: Addressing unemployment and poverty in urban areas.**
- **Features: Integration of economic programs (job creation) and social programs (combating marginalization).**
- **Advantage: Realization that urban planning alone is not enough to fix the city.**
- **Disadvantage: Weak financial sustainability, with some cities relying solely on government support.**

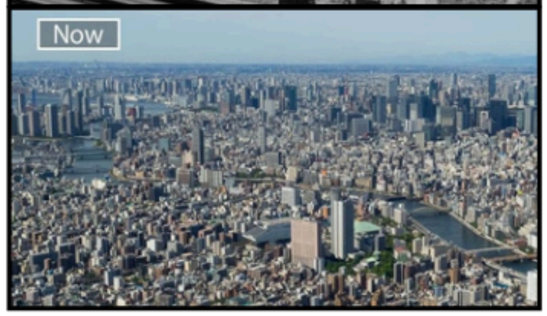
d) Sustainability and Participatory Phase (1990–Present)

- **Focus: Sustainable development, improving quality of life, and community participation.**
- **Features: Environmentally friendly projects, use of renewable energy, and preservation of heritage.**
- **Advantage: Achieving a balance between the environment, economy, and society.**
- **Disadvantage: Difficulty in implementation in poor or rapidly growing cities.**













URP 233 2025	Team Work	Theoretical background	Urban Regeneration of Al-Ruwais District in Jeddah	

URBAN REGENERATION

Theoretical Foundations

- Functionalist Theory: The city is a system of functions (housing, work, leisure), and when these functions fail, deterioration occurs.
- Sustainable Development Theory: Focuses on achieving a long-term balance between the environment, the economy, and society.
- Participatory Governance Theory: Emphasizes the importance of involving all stakeholders (governments, private sector, local community).

Advantages

- Revitalization of old urban centers.
- Improvement of quality of life and services.
- Job creation and economic stimulation.
- Protection of architectural heritage and reuse of historic buildings.
- Pollution reduction and environmental improvement.

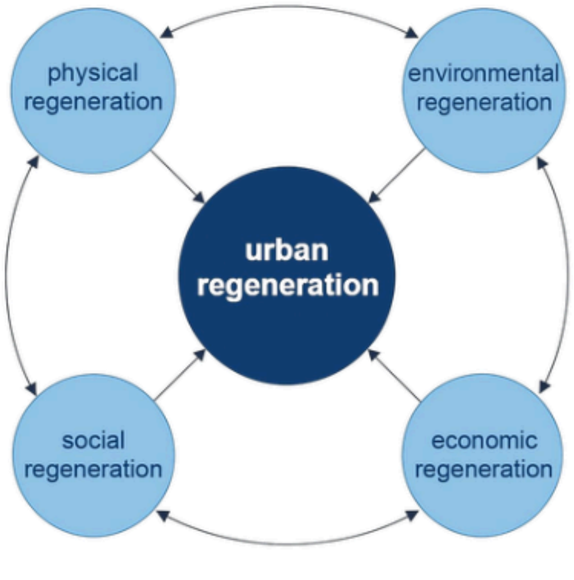
Challenges and Disadvantages

- Displacement: Rising housing prices lead to the expulsion of original residents.
- High Financial Cost: Requires massive investments.
- Administrative Complexity: Difficulty in coordination between various stakeholders.
- Project Failures: May occur in the absence of public participation.

Dimensions of Urban Renewal

- Urban Dimension: Improving the built environment and rehabilitating infrastructure.
- Economic Dimension: Supporting commercial activities and creating new job opportunities.
- Social Dimension: Promoting social justice and providing affordable housing.
- Environmental Dimension: Reducing pollution and increasing green spaces.
- Cultural Dimension: Preserving heritage and enhancing local identity.











URP 233 2025	Team Work	Theoretical background	Urban Regeneration of Al-Ruwais District in Jeddah

URBAN REGENERATION

Overview of the Area

The Financial District in Manhattan is considered one of the most important financial centers in the world. Over time, however, the area began to experience a decline in both residential and commercial activity, particularly during the late twentieth century, as it had been heavily dependent on business activities limited to working hours. This decline prompted the launch of a comprehensive urban regeneration process to transform the district into a mixed-use environment.

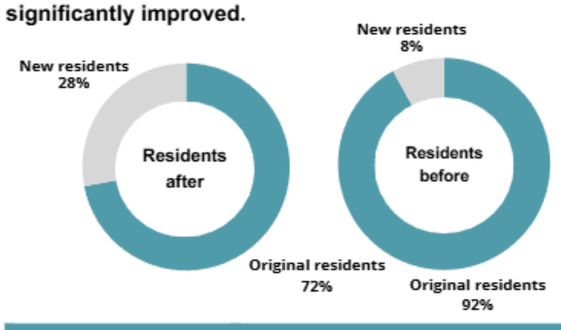
Before regeneration:

The district suffered from a set of fundamental challenges. The most critical was the very limited residential population, which reached only about 25,920 people in the year 2000. In addition, most commercial spaces became nearly vacant after working hours, which caused a 30% drop in overall sales volume.

After regeneration:

Numerous residential towers and commercial projects were developed, contributing to a 28% increase in the population, raising the number of residents to 36,726. Commercial activity also grew by 45% due to the attraction of new investments. Before regeneration in 2000, the area recorded only about 250 million USD in annual commercial revenues, while by 2020, revenues had surged to 365 million USD per year.

In addition, the district's green spaces were significantly improved.



Environmental dimension

- Air quality improved by 37% (2009–2021) due to pollution reduction, and about 10 hectares of new green spaces were created in Battery Park.
- The city responded with major projects such as the Lower Manhattan Coastal Resiliency (LMCR) to elevate waterfronts and establish green flood barriers.

Social dimension

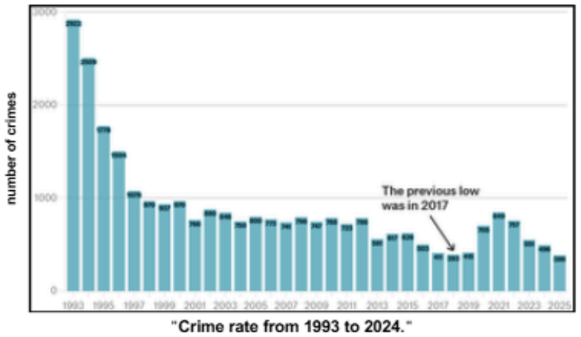
- Approximately 141 hectares of office space have been converted into housing, most of it in the Financial District, up to the year 2025.
- The crime rate decreased by 82% between 1993 and 2024, making the area significantly safer .

Urban dimension

- Fulton Center (2014)
- WTC Transportation Hub (2016)
- Redevelopment and rehabilitation of historic buildings, converting them into new uses.
- Enhancement of waterfronts and public streets to promote walkability and tourism.

Economic dimension

- Manhattan attracted about 63.3 million visitors in 2023, a figure close to pre-pandemic levels (66.6 million in 2019).
- The Financial District stood out as a major destination through Wall Street and the World Trade Center site.
- Average rents increased significantly, with some neighborhoods experiencing a rise of +43% between 2021–2023.



"Crime rate from 1993 to 2024."



Top View Financial District(2001)



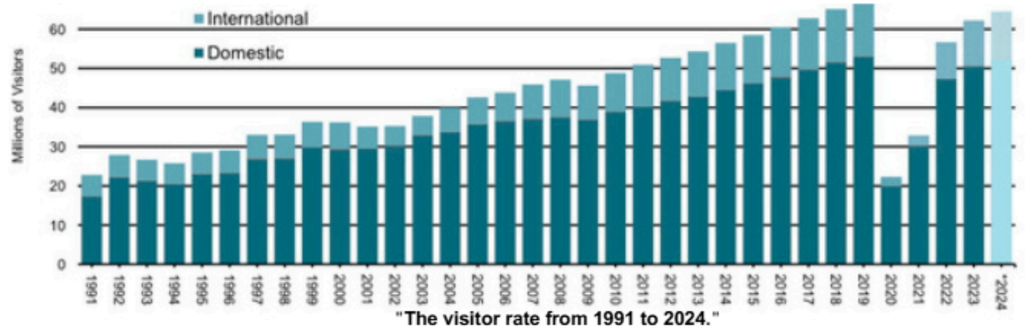
Top View Financial District (2020)



WTC Transportation



Lower Manhattan Coastal Resiliency (LMCR)



"The visitor rate from 1991 to 2024."

URP 233 2025	Team Work	Best Practice NewYork City (Manhattan)	Urban Regeneration of Al-Ruwais District in Jeddah

URBAN REGENERATION

Introduction

Al-Ruwais Neighborhood is one of the oldest districts in Jeddah. It is located in the heart of the city, on the northern side, and serves as a meeting point between vibrant commercial and residential centers.

The population of Al-Ruwais is estimated at around 40,000 residents, including approximately 6,440 Saudi citizens (16%) and 32,560 expatriates. The neighborhood covers an area of about 4 km².

Importance of the Neighborhood

the location of the neighborhood is situated near important districts such as Al-Hamra and Al-Sharafiyah, and is intersected by the vital Hail Road, which connects it to various parts of the city.

Its services: the neighborhood contains numerous facilities, including schools, health centers, shopping centers and markets, as well as a variety of cafés and restaurants.

Main Problems of the Neighborhood

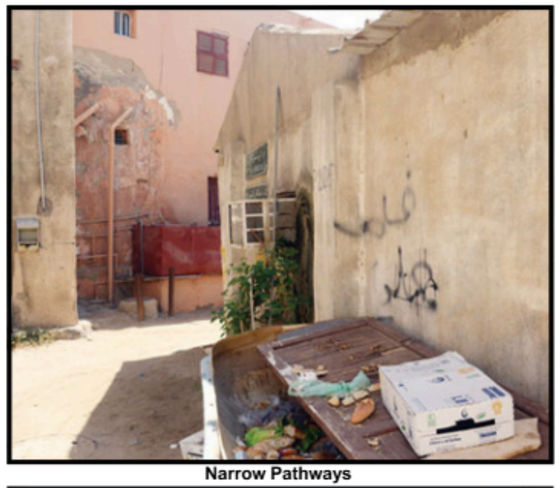
Deterioration of buildings: Many dilapidated and collapse-prone structures exist, with more than 800 buildings identified as being in a severe state of deterioration.

Infrastructure: The neighborhood faces challenges in sewage systems, electricity, and congestion in certain areas.

Services: Reports showed that internet services, entertainment facilities, and infrastructure (such as roads and utilities) require improvement. The neighborhood received a score of 54.27% for street and facility quality, 87% for internet, 85% for medical services, and 75% for educational services, while electricity and water received around 85%



Buildings Condition



Narrow Pathways



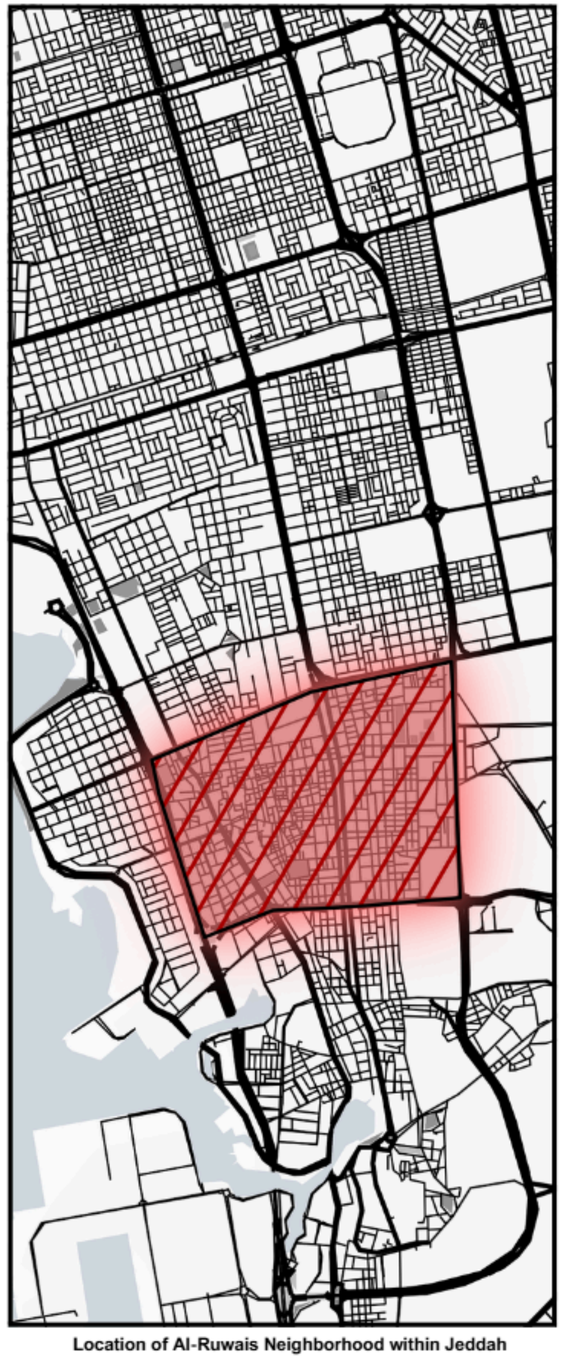
Aerial View



Bad Street Network

Vision of Regeneration

- Reposition Al-Ruwais as a vital urban core that reflects Jeddah's historical identity and future ambitions.
- Transform the physical environment into a livable, inclusive, and sustainable urban setting.
- Preserve and reinterpret the neighborhood's cultural heritage through contemporary urban design.
- Encourage social and economic renewal by enhancing accessibility, mixed-use diversity, and public life.
- Promote environmental sustainability and resilience through climate-responsive planning and green infrastructure.



Location of Al-Ruwais Neighborhood within Jeddah

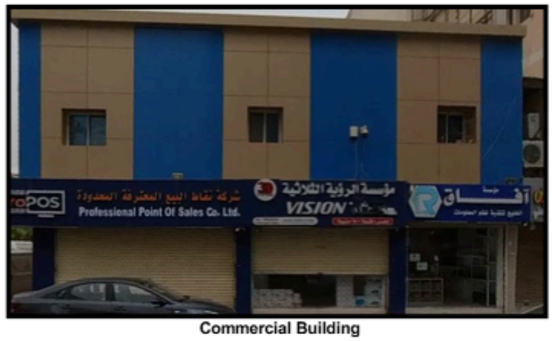
URBAN REGENERATION



King Saud Mosque



Bangladesh international school

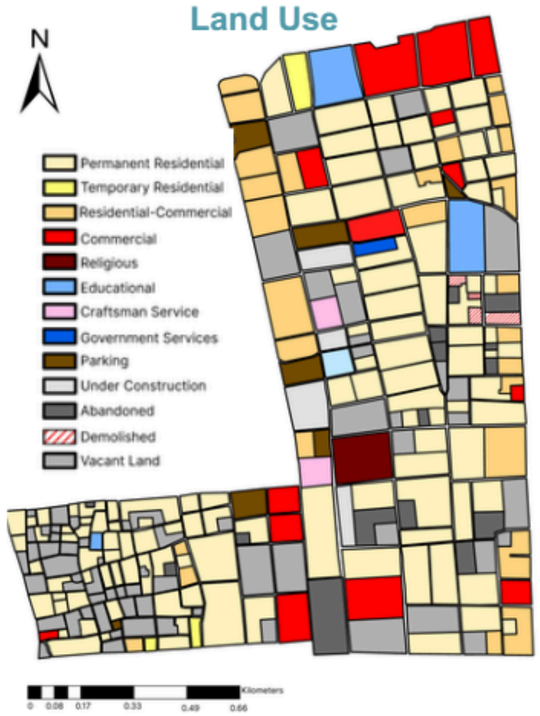


Commercial Building

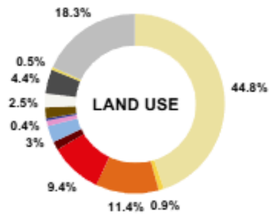


Residential Building

Land Use



The pie chart shows that permanent residential use dominates the area, followed by vacant land, while other uses such as commercial, service, and educational occupy smaller proportions.



Land Use	Area (Hectare)
Permanent Residential	34.76
Temporary Residential	0.72
Residential-Commercial	8.82
Commercial	7.31
Religious	1.17
Educational	2.33
Craftsman Service	0.77
Government Services	0.32
Parking	0.91
Under Construction	1.94
Vacant Lands	3.39
Demolished	0.37
Vacant Land	14.82352



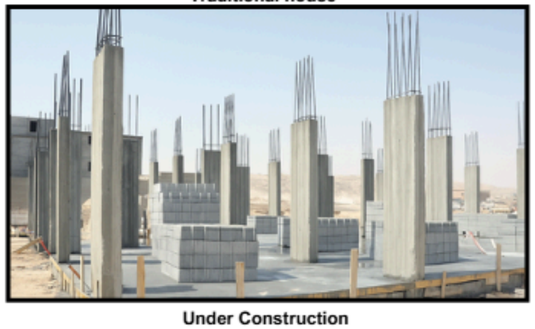
Apartment



Villa



Traditional house

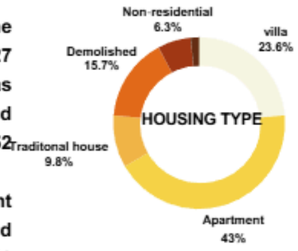


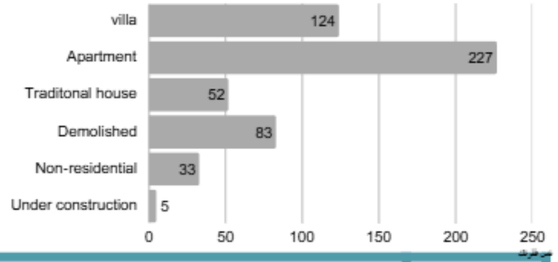
Under Construction

Housing Type



Apartments dominate the area, making up 43% (227 buildings), followed by villas at 23.6% (124 buildings) and traditional houses at 9.8% (52 buildings).
Demolished buildings account for 15.7% (83 buildings) and non-residential ones for 6.3% (33 buildings), with only a few under construction, indicating limited ongoing development.

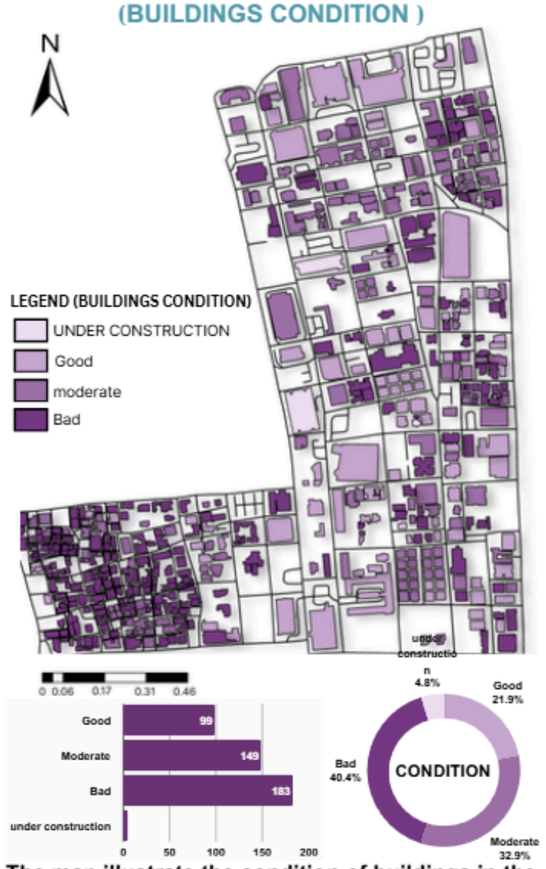




URP 233 2025	Team Work	Current Situation Analysis of Al-Ruwais District	Urban Regeneration of Al-Ruwais District in Jeddah

URBAN REGENERATION

(BUILDINGS CONDITION)



The map illustrate the condition of buildings in the study area, showing a predominance of structures in bad and moderate states. Buildings in bad condition represent the largest share at 40%, followed by those in moderate condition at 32.9%, while only 21.9% are in good condition.



BAD CONDITION

(BUILDINGS AGE)

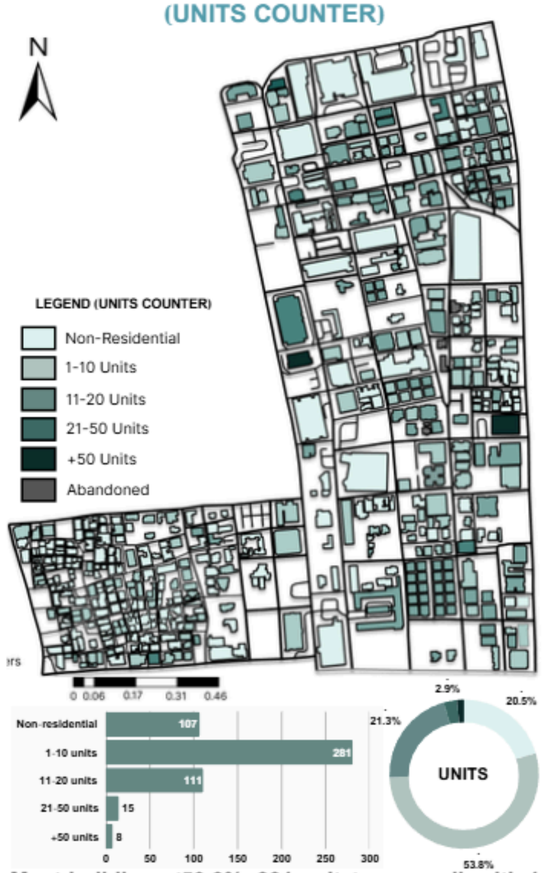


The map indicates that 80.5% of the buildings in the area are old, indicating a lack of modern development. Only 3.7% of the buildings are new, suggesting weak urban growth. There is an urgent need for infrastructure renewal and increased investment to improve quality of life.



OLD BUILDING

(UNITS COUNTER)

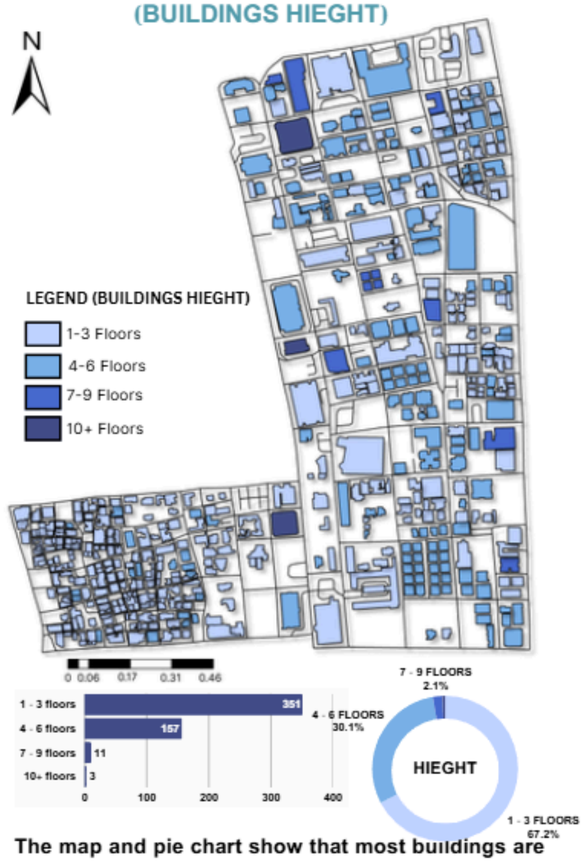


Most buildings (53.8%, 281 units) are small, with 1–10 residential units. Mid-size buildings (11–20 units) form 21.2% (111), and non-residential ones 21.3% (107). Only 2.9% (15) are large. Overall, the area is mainly low-density residential with few multi-unit structures.



MULTI-UNIT HOUSING

(BUILDINGS HIEGHT)



The map and pie chart show that most buildings are low-rise, with 1–3 floors representing about 67.2% of the total. Medium-rise buildings (4–6 floors) come next at 30.1%. Buildings with 7–9 floors account for only 2.1%, and those over 10 floors are very rare, indicating potential for future vertical growth and more efficient land and infrastructure use.

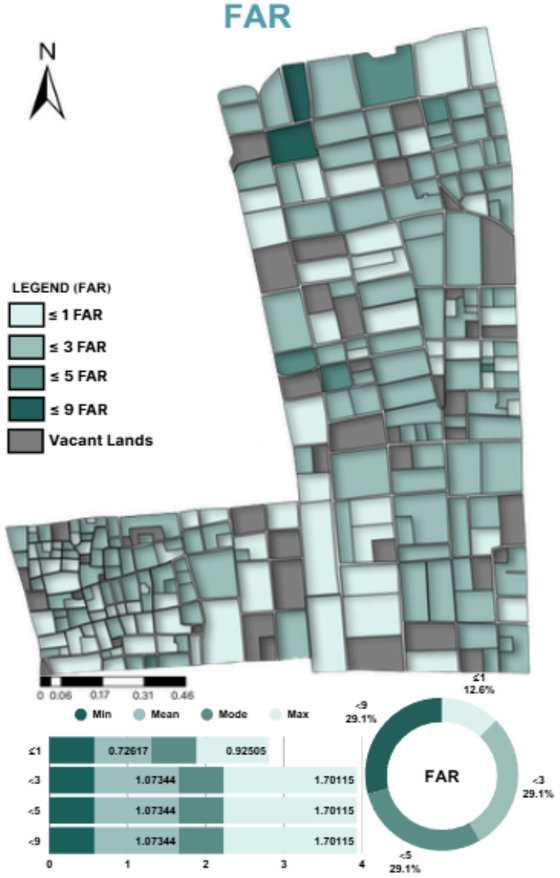


BETWEEN 1 AND 6

URP 233 2025	Team Work	Current Situation Analysis of Al-Ruwais District	Urban Regeneration of Al-Ruwais District in Jeddah

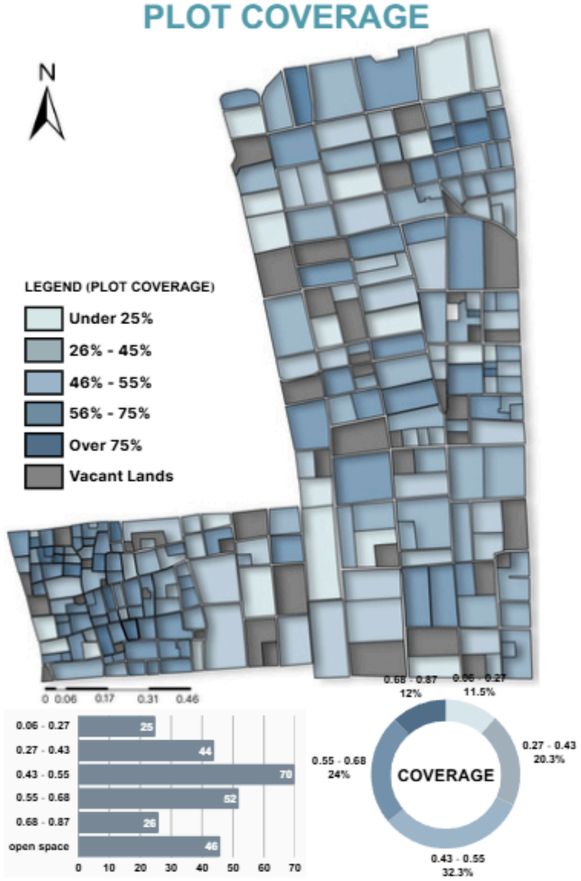
URBAN REGENERATION

FAR



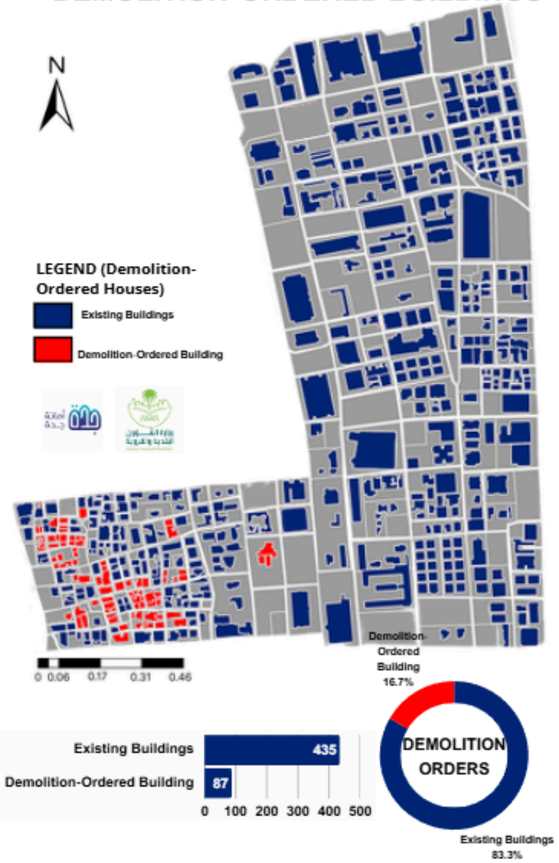
The maps and charts show FAR distribution in Al-Ruwais and Al-Sharafiyah, mostly low to medium density (0.7–0.8), with few outliers along main streets. The pattern reflects a low-rise character with potential for future densification.

PLOT COVERAGE



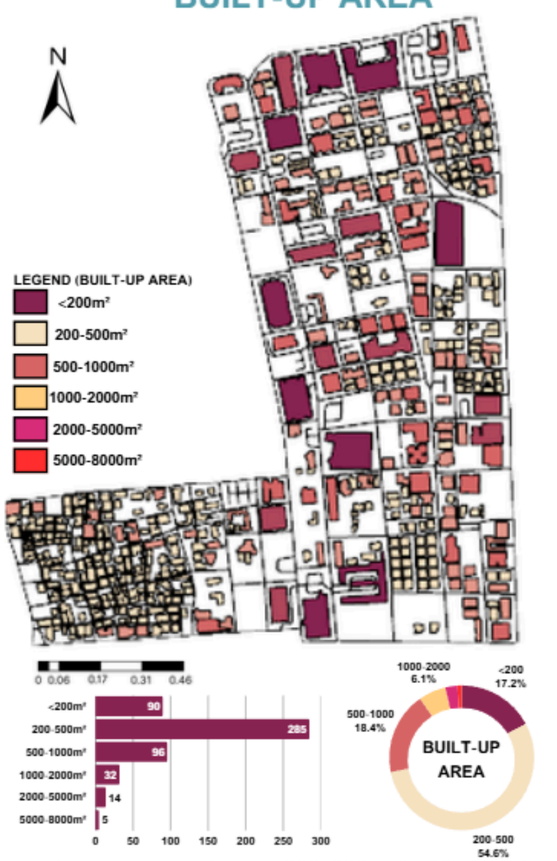
Most values fall between 0.4 and 0.6, making up the largest share. Values above 0.8 are very rare. The distribution is slightly right-skewed, showing a concentration at lower values with only a few high extremes, indicating limited variability and moderate intensity across most of the dataset.

DEMOLITION-ORDERED BUILDINGS



Out of the total buildings on site, 83.3% (435 buildings) are existing and in use, while 16.7% (87 buildings) are marked for demolition This indicates that most of the built environment is structurally stable, with limited areas requiring redevelopment.

BUILT-UP AREA



Most plots fall between 200–500 m² (55%), with smaller shares under 200 m² (17%) and 500–1000 m² (18%). Larger built-up areas above 1000 m² are limited, and very large ones (>5000 m²) are rare. The distribution reflects a fine-grained urban fabric with few large outliers.



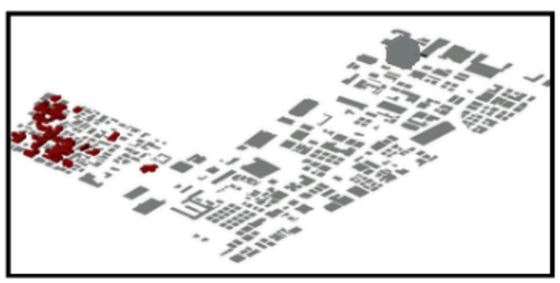
HIGH FLOOR AREA RATIO



HIGH PLOT COVERAGE



LOW PLOT COVERAGE



DEMOLITION-ORDERED BUILDING



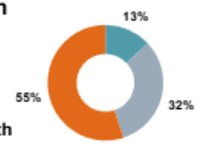
SMALL BUILT-UP PLOTS

URP 233 2025	Team Work	Current Situation Analysis of Al-Ruwais District	Urban Regeneration of Al-Ruwais District in Jeddah

URBAN REGENERATION

Road Condition

Satisfied
Partially Satisfied
Unsatisfied

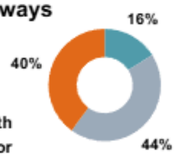


In Al-Ruwais, 55% are unsatisfied with roads, showing poor condition and .maintenance



Sidewalks & Walkways

Satisfied
Partially Satisfied
Unsatisfied

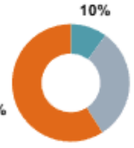


In Al-Ruwais, only 16% are satisfied with sidewalks, while most find them poor or .average, needing repair and improvement



Traffic

Satisfied
Partially Satisfied
Unsatisfied

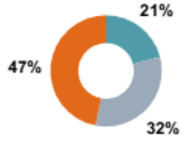


In Al-Ruwais, traffic satisfaction is very low — only 10% satisfied, while 59% unsatisfied, showing .serious congestion problems



Parking

Satisfied
Partially Satisfied
Unsatisfied

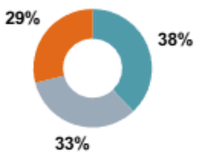


In Al-Ruwais, parking satisfaction is low — only 21% satisfied, while 47% unsatisfied, .showing a clear lack of parking spaces



Cleaning Services

Satisfied
Partially Satisfied
Unsatisfied

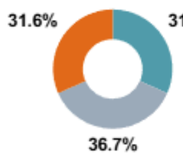


In Al-Ruwais, satisfaction is mixed: 38% satisfied, 33% .partial, 29% unsatisfied



Building Condition

Satisfied
Partially Satisfied
Unsatisfied

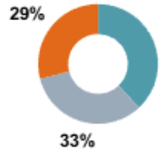


In Al-Ruwais, building condition is moderate — 31.6% satisfied, 31.6% .unsatisfied



Service

Satisfied
Partially Satisfied
Unsatisfied

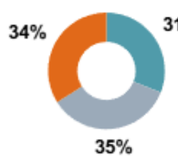


In Al-Ruwais, service satisfaction is low — only 19% satisfied and 48% unsatisfied, .showing need for better services



Level of Safety

Satisfied
Partially Satisfied
Unsatisfied

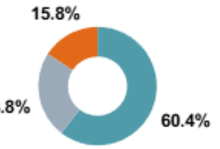


In Al-Ruwais, safety levels are moderate — 31% satisfied, 35% partially satisfied, .and 34% unsatisfied



Mosque

Satisfied
Partially Satisfied
Unsatisfied

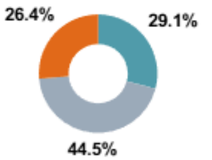


Features satisfaction data showing 60.4% Satisfied. Also includes photos of large, .white mosques in an urban setting



Educational Services

Satisfied
Partially Satisfied
Unsatisfied

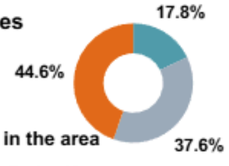


Educational Services slide: 44.5% Partially Satisfied in the survey; contrasts a modern .building photo with an old gate



Health Services

Satisfied
Partially Satisfied
Unsatisfied



.There are no healthcare services in the area

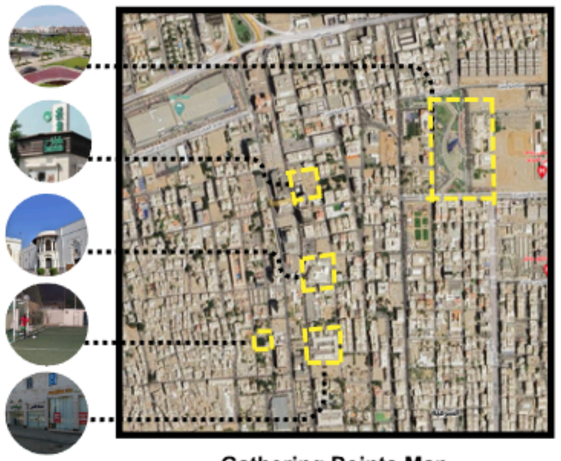
Health Services: 44.6% Unsatisfied. Key .finding: No healthcare services in the area

Citizens' Opinions

.Healthcare services are far; we need closer facilities
Parking spaces are insufficient, causing constant congestion
.The district needs to be rebuilt from scratch

Residents' Opinions

.There is a lack of green spaces and recreational areas
.Sidewalks are not suitable for walking, especially for the elderly
Street lighting is weak in some places, making us feel unsafe at" ".night



Gathering Points Map



Social Survey Pattern Map

The two maps show survey locations and residents' satisfaction. The upper and lower areas have low satisfaction due to poor services, while the central areas show mixed responses, highlighting social .and spatial differences in service quality

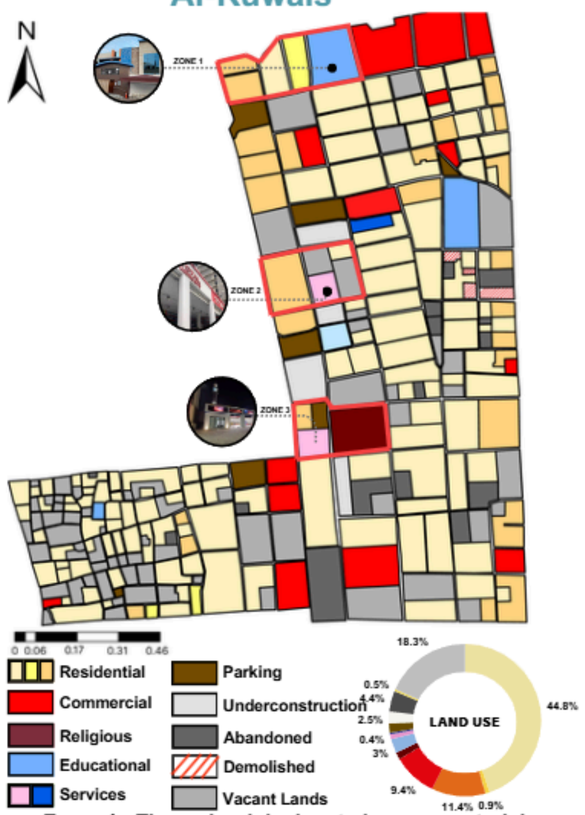
Weaknesses 🔴 Strengths 🟢 Overall Insight 🟡

Overall satisfaction is moderate, but poor roads, traffic, parking, and sidewalks show major infrastructure issues. Improvements should focus on roads, parking, and walkways to enhance .satisfaction

URP 233 2025	Team Work	Satisfaction Rates of Services	Urban Regeneration of Al-Ruwais District in Jeddah

URBAN REGENERATION

Al-Ruwais



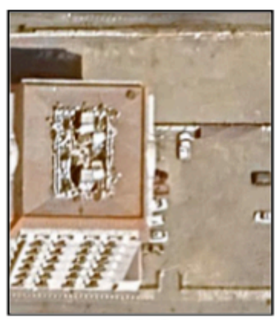
- Zone 1: The school is located on an arterial road, which poses a risk to students and parents.
- Zones 2 and 3: Car workshops are located in the middle of residential buildings, causing odors, congestion, and disturbance to residents.

Land Use Type	Al-Ruwais	Al-Jamiah "Al Qaswa" (Benchmark)
Residential	56.89% (44.3 HA)	47.0% (43.05 HA)
Educational	2.99% (2.33 HA)	8.2% (7.34 HA)
Religios	1.50% (1.17 HA)	3.3% (2.94 HA)
Commercial	9.3% (7.31 HA)	7.7% (6.87 HA)
Services	1.40% (1.09 HA)	4.3% (3.82 HA)
Green Areas	0% (0 HA)	20.0% (26.00 HA)

Solid And Void



- High void ratio in the north — leads to underutilized land and weak urban continuity.
- Dense southern area — overcrowded built-up blocks reduce ventilation and visual comfort.
- Lack of transition between solid and void , no gradual or planned shift between built and open areas.



SOLID



VOID

Open Spaces



- Lack of greenery and functional use — most open spaces are parking or vacant land, not landscaped or shaded
- Poor quality and maintenance — visible barriers, rough surfaces, and neglected areas reduce usability.
- Low accessibility and safety many spaces are enclosed or uninviting, discouraging public activity.

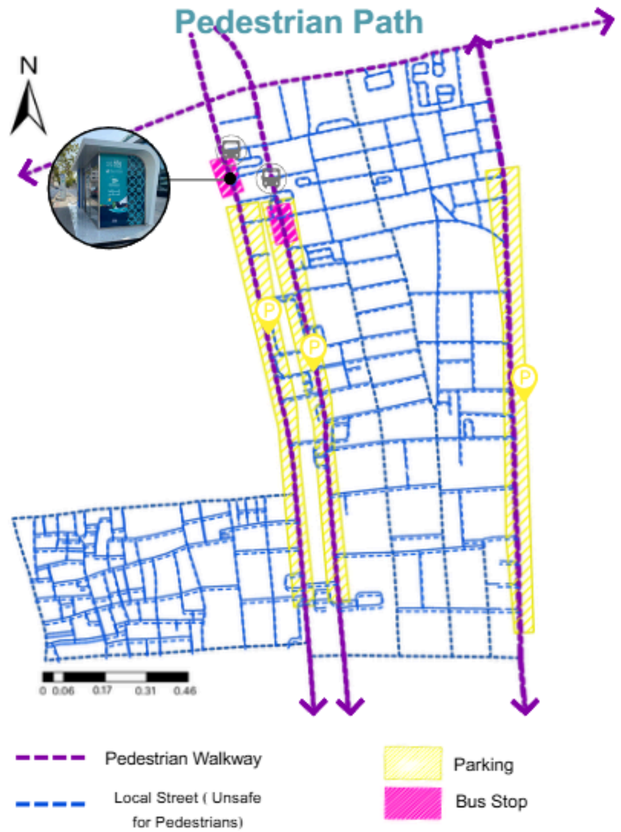


VACANT LAND



PARKING

Pedestrian Path



- Discontinuous walkways — pedestrian paths are fragmented and lack full connectivity.
- Limited shaded or safe areas — absence of trees, canopies, or buffer zones from vehicles.
- Dominance of vehicular movement — wide streets and parking reduce pedestrian priority.



PEDESTRIAN PATH - AL-MADINA ROAD



LACK OF PEDESTRIAN PATH

URP 233 2025	Team Work	Urban Analysis	Urban Regeneration of Al-Ruwais District in Jeddah